

sors to identify, grasp and manipulate parts. Some assembly robots can also handle assembly of parts that are too small for human hands to handle them accurately.

5. AI and Quality 4.0

The core concept of Quality 4.0 aligns quality management with Industry 4.0 techniques. Technological advances in Industry 4.0 such as advanced analytics, AI and machine learning can help organisations achieve real-time visibility of vital quality metrics, supplier performance, and customer services.

Due to the rising complexity in the products and the spare parts, manufacturers may face the challenge to maintain high-quality products. Quality 4.0 ensures improved quality of products and enhanced output efficiency by using AI algorithms. If any kind of issues are found at the initial stages, they can be taken care of immediately.

6. Predictive analytics

AI, machine learning and IoT are used in modern manufacturing industries. Here, data is one of the most important assets. Predictive analytics tools are used to analyse the incoming data to identify problems in advance. AI-powered predictive analytics promises to revolutionise analytics. Predictive tools give manufacturers the power to analyse massive amounts of data and discover correlations between critical variables, enabling them to address the root causes of problems before quality issues occur. Predictive quality is one of the most important use cases in manufacturing. Predictive maintenance makes use of AI algorithms to predict failures of a system or machine to prevent failures and prevent downtimes in the manufacturing units.

With advanced data analysis and real-time tracking, machinery maintenance can be scheduled before any problem arises. These technologies help drive operational efficiency, reduce the manufacturing operating expenditure and improve productivity.

7. AI and generative designs

A designer in the manufacturing industry first establishes a request-for-quotation, provides the design assets, indicates the processes required, such as dimensions, materials, volume and

turnaround time. Depending on the product, the actual designing phase involves lots of imagination and creativity of the designer.

With the generative design concept, a designer is no longer limited by his own imaginations or past experience. The designer just needs to input design parameters (such as materials, size, weight, strength, manufacturing methods, and cost constraints) into generative design software, and the AI algorithm explores all the possible combinations of a solution, generates many or thousands of design options. The designer can filter from the generated design options and select the best options that would meet his requirements.

8. AI and cost optimisation

One of the major benefits of AI in manufacturing is cost optimisation. AI enables manufacturers to cost-effectively create and maintain their own algorithm in-house, which is cheaper, more versatile, and more adaptive to constantly changing market conditions. AI can fully automate complex tasks, provide consistent and precise optimum set points. It requires less manpower to maintain and adjust quickly when management revises manufacturing strategy and production plans.

In some cases, AI can partially or even fully replace certain employees and you need not include their salaries in your budget. AI is used to cut maintenance and repair costs.

9. Inventory optimisation

Inventory management plays an important role in manufacturing because it can make or break a business. One item out of stock can bring a business to its knees. Many companies run into losses due to bad inventory management. Unlike inventory management in retailers and wholesalers, which consists items ready for sale, a manufacturing company's inventory will include goods in various stages of production, from raw materials to products ready to ship to customers. A typical manufacturer uses three types of inventories: raw materials, work in process and finished goods. Raw materials including steel, wood, plastic, chemicals, etc, are the main inputs of production. Work in

process represents goods that are still in the production process. Finished goods are those that are awaiting sale or ready to ship to customers.

AI in inventory provides many benefits for the companies that are utilising it. For example, in the car manufacturing industry, smart warehouses are inventory systems, where the inventory process is partially or entirely automated using modern technologies such as AI, machine learning and Industrial Internet of Things (IIoT). These technologies enable inventory systems to connect each process, collect data at each of the nodes, continuously learn and optimise the process and increase productivity. AI and machine learning models are able to scale inventory optimisation across all distribution locations and independent variables that affect demand and supply performance.

10. AI and blockchain

Many products are being manufactured globally but do you know from where these products originated, how and when they were produced and shipped to the buyers? The demand for transparency in the manufacturing supply chain has significantly increased in today's global market. Blockchain can be used in manufacturing and perfectly fits for this solution to track manufactured goods through the supply chain.

Blockchain involves a digital ledger that is continuously updated to record and track transactions, accounting and asset movement. It is new digital information, which stores data in an encrypted, distributed ledger format. AI can manage blockchain efficiently because AI and encryption work very well together. In manufacturing, blockchain can be used to establish an organised digital thread, track the history of a part, from its digital design to production all the way to end of life. This digital data can be shared with multiple parties that get access to the same information.

Some of the advantages of blockchain include monitoring supply chain for greater transparency, materials provenance and counterfeit detection, engineering design, identity management, asset tracking, quality assurance and regulatory compliance. **EFY**